SATBAYEV UNIVERSITY

CSE6031 | Mathematics & Statistics | Computer Science

LABORATORY WORK 5

The Normal Distribution and Central Limit Theorem

Normal Probabilities, z-Scores and Sampling Distributions

Short theory. The Normal distribution uses mean and standard deviation to model continuous variation. The CLT explains why sample means become approximately Normal under broad conditions.

Learning objectives

- Use z-scores and Normal probabilities
- Calculate standard error
- Demonstrate the CLT computationally

Main task

Individual API response times have mean 120 ms and standard deviation 30 ms. Assume the population is not necessarily Normal.

Tasks:
1. For a Normal model, calculate the z-score for 150 ms and P(X<150).
2. Draw samples of size n=36. According to the CLT, describe the approximate sampling distribution of the sample mean.
3. Calculate the standard error.
4. Calculate P(sample mean > 130 ms).
5. Simulate 20,000 samples and compare the simulated probability with the CLT result.
6. Repeat with n=100 and explain how sample size changes the standard error and probability.
7. Plot the sampling distributions for n=36 and n=100.

Python implementation

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

mu, sigma = 120, 30

# Individual observation under a Normal model
z = (150-mu)/sigma
```

```
print("z for 150 ms:", z)
print("P(X<150):", norm.cdf(z))

for n in [36, 100]:
    se = sigma/np.sqrt(n)
    theory = 1 - norm.cdf(130, loc=mu, scale=se)

    # Use a skewed population with the same approximate mean and SD
    # Gamma(shape=16, scale=7.5) has mean 120 and SD 30
    samples = np.random.gamma(shape=16, scale=7.5, size=(20000,n))
    means = samples.mean(axis=1)
    sim = np.mean(means > 130)

    print("\nn =", n)
    print("SE =", se)
    print("CLT P(mean>130) =", theory)
    print("Simulation =", sim)

    plt.figure()
    plt.hist(means, bins=35, density=True)
    xx = np.linspace(means.min(), means.max(), 300)
    plt.plot(xx, norm.pdf(xx, loc=mu, scale=se))
    plt.xlabel("Sample mean response time")
    plt.ylabel("Density")
    plt.title(f"Sampling Distribution of Mean, n={n}")
    plt.show()
```

Required submission

- Mathematical/statistical calculation where applicable
- Working Python code
- Required numerical output and graph(s)
- 3-5 sentence interpretation connected to the Computer Science scenario

Key takeaway: The CLT concerns the distribution of sample means. Its standard error is sigma/sqrt(n), so larger samples produce more precise sample means.

Assessment

Criterion	Points
Correct method/model	2
Mathematical result	2
Python code	2
Visualization/output	2
Interpretation	2